

KENDRIYA VIDYALAYA SANGATHAN
SILCHAR REGION
PRE-BOARD EXAMINATION
SESSION 2024-25

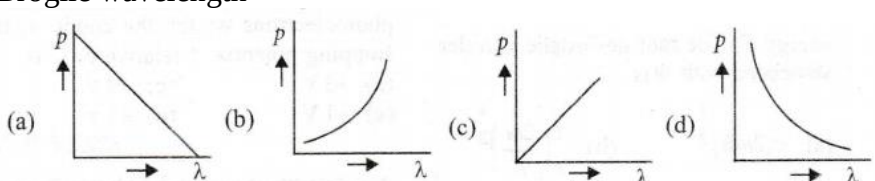
CLASS: XII
SUB: PHYSICS

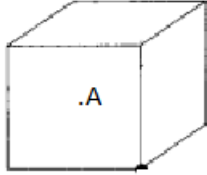
M.M.: 70
TIME: 3 HOURS

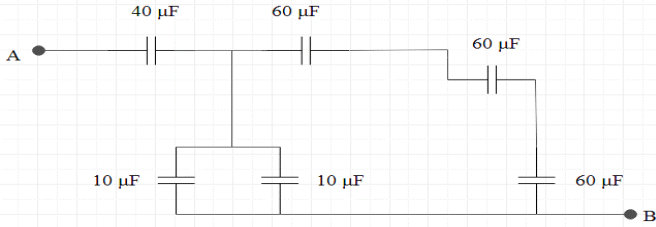
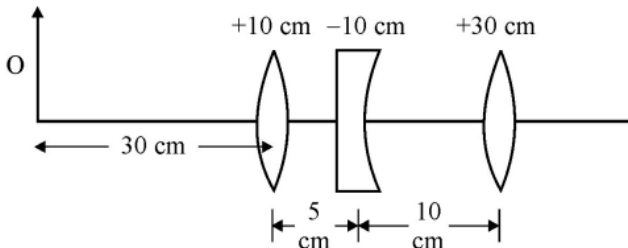
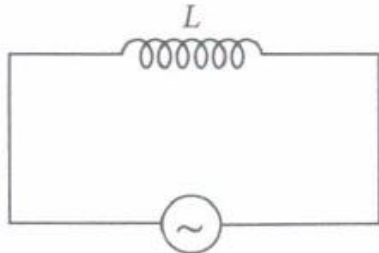
General Instructions:

- There are 33 Questions in all. All questions are compulsory.
- This Question paper has 5 sections; Section A, Section B, Section C, Section D and Section E. All sections are compulsory
- Section A** contains **16 questions - 12 MCQs and 04 Assertion-Reasoning** based of 1 mark each, **Section B** contains **5 questions of 2 marks each**, **Section C** contains **7 Questions of 3 Marks each**, **Section D** contains **2 Case study Based Questions of 4 marks each** and **Section E** contains **3 Long Answer Questions of 5 marks each**.
- There is no overall choice. However, an internal choice has been provided in 1 question in Section B, 1 question in Section C, 1 question in each CBQ in Section D and all three questions in Section E. You have to attempt only one of the choices in such questions.
- Use of calculators is not allowed.
- You may use the following values of physical constant wherever necessary.
 - $c = 3 \times 10^8$ m/s
 - $m_e = 9.1 \times 10^{-31}$ Kg
 - $m_p = 1.7 \times 10^{-27}$ Kg
 - $e = 1.6 \times 10^{-19}$ C
 - $\mu_0 = 4\pi \times 10^{-7}$ TmA⁻¹
 - $h = 6.63 \times 10^{-34}$ Js
 - $\epsilon_0 = 8.854 \times 10^{-12}$ C²/Nm²
 - Avogadro's Number = 6.023×10^{23} per gram mole

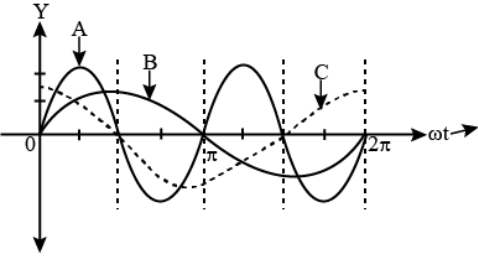
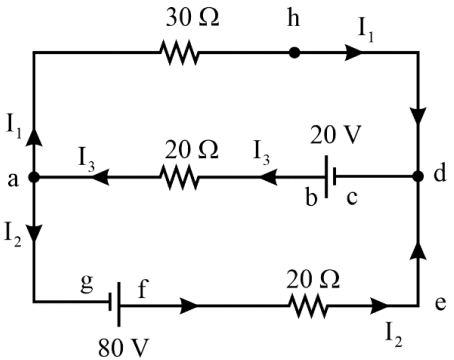
S.N	SECTION A	M
1	Which of the following is NOT the property of equipotential surface? (a) They do not cross each other. (b) The rate of change of potential with distance on them is zero. (c) For a uniform electric field they are concentric spheres. (d) They can be imaginary spheres.	1
2	An electron is moving along positive x-axis in a magnetic field which is parallel to the positive y-axis. In which direction will the magnetic force be acting on the electron? (a) Along -x axis (b) Along -z axis (c) Along +z axis (d) Along -y axis	1
3	Resistances P, Q, S and R are arranged in a cyclic order to form a balanced Wheatstone's network. The ratio of power consumed in the branches (P+Q) and (R+S) is (a) 1:1 (b) R:P (c) P ² :Q ² (d) P ² :R ²	1
4	A thin converging lens of focal length 10 cm and a thin diverging lens of focal length 20 cm are placed coaxially in contact. The power of the combination is: (a) -5D (b) +5 D (c) +15 D (d) - 15D	1

5	If a magnetic material is having magnetic susceptibility (χ) = -1, then the relative magnetic permeability (μ_r) and type of magnetic material will be (a) 2, Ferromagnetic (b) 1, Paramagnetic (c) 0, Diamagnetic (d) -1, diamagnetic	1
6	When alpha particles are bombarded on thin gold foil, most of them go straight through the foil because: (a) Alpha particles are much lighter than electrons (b) Alpha particles are positively charged (c) Most of the atom is empty space (d) Alpha particles move with high velocity	1
7	Which of the following statements is incorrect for the depletion region of a diode? (a) There the mobile charges exist. (b) There the mobile charges do not exist. (c) Recombination of holes and electrons take place in this region (d) None of these	1
8	A biconvex lens of glass having refractive index 1.47 is immersed in a liquid. It becomes invisible and behaves as a plane glass plate. The refractive index of the liquid is (a) 1.47 (b) 1.62 (c) 1.33 (d) 1.51	1
9	Which of the following figure represents the variation of particle's momentum and its associated de-Broglie wavelength 	1
10	The radius of nth orbit r_n in terms of Bohr radius (a_0) for a hydrogen atom is given by the relation (a) na_0 (b) $\sqrt{n} a_0$ (c) $n^2 a_0$ (d) $n^3 a_0$	1
11	What is the geometric shape of the wavefront that originates when a plane wave passes through a convex lens? (a) Converging and spherical (b) Diverging and spherical (c) Plane (d) None of the above	1
12	What is the phase difference between electric and magnetic fields in an electromagnetic wave? (a) 0 (b) π (c) $\pi/2$ (d) $\pi/4$	1
Question No. 13 to 16 consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. Both A and R are false		
13	Assertion: When two charges q_1 and q_2 are kept at some distance apart, force acting between these charges is F. If a third charge q_3 is placed quite close to q_2 , the force will remain F Reason: The force acting between two charges does not depend upon the presence of any other charge	1
14	Assertion: When a piece of non-metal and a metal are dropped from the same height near the surface of the earth, the non-metallic piece will reach the ground first. Reason: Induced current in metal will increase the acceleration.	1
15	Assertion: Coloured spectrum is observed when we look through a muslin cloth.	1

	Reason : It is due to diffraction of white light on passing through fine slits	
16	Assertion: In photoelectric effect the kinetic energy of the emitted photoelectrons increases with increase in the intensity of incident light. Reason: Photoelectric current depends upon the wavelength of the incident light.	1
	SECTION B	
17	Calculate the radius of curvature of an equi-concave lens of refractive index 1.5, when it is kept in a medium of refractive index 1.4, to have a power of $-5D$?	2
18	Find the total flux through each face of the cube with side of length 'a' if a charge q is placed at the centre A of the cube.	2
		
19	Calculate the binding Energy of an alpha (α) particle in Mev from the following data : Mass of Helium Nucleus = 4.00260 u , Mass of neutron = 1.008662 u , Mass of proton = 1.007825 u, 1 atomic mass unit (1u) = 931 MeV	2
20	Plot a graph showing variation of de Broglie wavelength (λ) associated with a charged particle of mass m, versus $1/\sqrt{V}$, where V is the potential difference through which the particle is accelerated. Find the de Broglie wavelength of an electron moving with a kinetic energy of 100 eV.	2
21	The coil of an ac generator consists of 100 turns of wire, each of area 0.5 m^2 . The resistance of the wire is $100\ \Omega$. The coil is rotating in a magnetic field of 0.8 T perpendicular to its axis of rotation, at a constant angular speed of 60 radian per second. Calculate the maximum emf generated and power dissipated in the coil. OR Derive the expression for the torque acting on a current carrying loop placed in a magnetic field.	2
	SECTION C	
22	A uniform conducting wire of length $12a$ and resistance R is wound up as a current carrying coil in the shape of (i) an equilateral triangle of side 'a' (ii) a square of sides 'a' and, (iii) a regular hexagon of sides 'a'. The coil is connected to a voltage source V_0 . Find the magnetic moment of the coils in each case. (area of hexagon = $\frac{3\sqrt{3}}{2} a^2$)	3
23	Derive the expression for electric field inside and outside for a charged spherical shell using Gauss theorem. Draw the graph showing the variation of electric field with distance from centre for this shell. OR Derive an expression for the electric field at any point on the equatorial line of an electric dipole.	3
24	The electric field in a plane electromagnetic wave is given by $E_z = 60 \sin(500x + 1.5 \times 10^{11}t)\text{V/m}$. (a) Write wavelength and frequency of the wave. (b) Write the expression of the magnetic field.	3
25	Draw the circuit diagram of a full wave rectifier and explain its working. Also, show the input and output waveforms.	3
26	Find the equivalent capacitance of the combination of capacitors between the points A and B.	3

			
27	(a) State briefly the processes involved in the formation of p-n junction explaining clearly how the depletion region is formed. (b) Draw the V-I characteristics of a p-n junction in (i) Forward bias and (ii) Reverse bias	3	
28	Three lenses of focal lengths +10 cm, −10 cm and +30 cm are arranged coaxially as in the figure given below. Find the position of the final image formed by the combination. 	3	
SECTION D (CASE STUDY BASE QUESTIONS)			
29	Let a source of alternating e.m.f $E = E_0 \sin \omega t$ be connected to a circuit containing a pure inductance L. If I is the value of instantaneous current in the circuit, then $I = I_0 \sin(\omega t - \pi/2)$. The inductive reactance limits the current in a purely inductive circuit and is given by $X_L = \omega L$ (A) A 100 hertz a.c. is flowing in a 14 mH coil. The reactance is (a) 15Ω (b) 7.5Ω (c) 8.8Ω (d) 10Ω (B) What is the reactance of an inductor in a DC circuit? (a) Maximum (b) Minimum (c) zero (d) Infinite (C) In a inductive circuit, by what value of phase angle does alternating current lags behind e.m.f.? (a) 45° (b) 90° (c) 120° (d) 75° (D) How much inductance should be connected to 200 V, 50 Hz a.c. supply so that a maximum current of 0.9 A flows through it? (a) 5 H (b) 1 H (c) 10 H (d) 4.5 H OR The maximum value of current when inductance of 2 H is connected to 150 volt, 50 Hz supply is (a) 0.337 A (b) 0.721 A (c) 1.521 A (d) 2.522 A		4
30	When light of sufficiently high frequency is incident on a metallic surface, electrons are emitted from the metallic surface. This phenomenon is called photoelectric emission. Kinetic energy of the emitted photoelectrons depends on the wavelength of incident light and is independent of the intensity of light. Number of emitted photoelectrons depends on intensity. $(h\nu - \Phi)$ is the maximum	4	

	kinetic energy of emitted photoelectrons (where Φ is the work function of metallic surface). Reverse effect of photo emission produces X-ray. X-ray is not deflected by electric and magnetic fields Wavelength of a continuous X-ray depends on potential difference across the tube. Wavelength of characteristic X-ray depends on the atomic number. (A) The number of photoelectrons emitted for light of frequency (higher than the threshold frequency ν_0) is proportional to: (a) threshold frequency (b) intensity of light (c) frequency of light (d) $\nu - \nu_0$ (B) The work-function for a metal is 3 eV. To emit a photoelectron of energy 2 eV from the surface of this metal, the wavelength of the incident light should be: (a) 6187 Å (b) 4125 Å (c) 12375 Å (d) 2875 Å (C) The stopping potential V_0 for photoelectric emission from a metal surface is plotted along with the y-axis and frequency ν of incident light along the x-axis. A straight line is obtained as shown. Planck's constant is given by (a) product of the slope of the line and charge on the electron (b) intercept along y-axis divided by the charge on the electron (c) product of the intercept along x-axis and mass of the electron (d) the slope of the line (D) What happens to the kinetic energy of the emitted electrons when the light is incident on a metal surface? (a) It varies with the frequency of light (b) It varies with the light intensity (c) It varies with the speed of light (d) It varies irregularly OR Two beams, one of red light and the other of blue light, of the same intensity are incident on a metallic surface to emit photoelectrons. Which emits electrons of greater frequency? (a) Both (b) Red light (c) Blue light (d) Neither	
	SECTION E	
31	Consider the experimental set up shown in the figure. This jumping ring experiment is an outstanding demonstration of some simple laws of Physics. A conducting non-magnetic ring is placed over the vertical core of a solenoid. When current is passed through the solenoid, the ring is thrown off. Answer the following questions: (i) Explain the reason of jumping of the ring when the switch is closed in the circuit. (ii) Explain what will happen if the terminals of the battery are reversed and the switch is closed? (iii) Explain the two laws that help us understand this phenomenon. OR (i) In a series LCR circuit connected across an AC source of variable frequency, obtain the expression for its impedance and draw a plot showing its variation with frequency of the AC source. (ii) What is the phase difference between the voltages across inductor and the capacitor at resonance in the LCR circuit ?	5

	(iii) A device 'X' is connected to an ac source $V = V_0 \sin \omega t$. The variation of voltage, current and power in one cycle is shown in the following graph : (a) Identify the device 'X'. (b) Which of the curves A, B and C represent the voltage, current and the power consumed in the circuit ? Justify your answer.		
32	Define the term polarization of a dielectric and write its relation with susceptibility. Use Kirchhoff's laws to determine the value of the current I_1, I_2 and I_3 flowing in the circuit shown below: OR An electric field is uniform and acts along + x direction in the region of positive x. It is also uniform with the same magnitude but acts in – x direction in the region of negative x. The value of the field is $E = 200 \text{ N/C}$ for $x > 0$ and $E = -200 \text{ N/C}$ for $x < 0$. A right circular cylinder of length 20 cm and radius 5 cm has its centre at the origin and its axis along the x-axis so that one flat face is at $x = +10 \text{ cm}$ and the other is at $x = -10 \text{ cm}$. Find : (i) The net outward flux through each flat face (ii) The flux through the side of the cylinder (iii) The net outward flux through the cylinder. (iv) The net charge present inside the cylinder.		5
33	(i) Draw the plot of intensity distribution in a diffraction pattern due to a single slit. (ii) Derive the expression for the intensity at a point of the interference pattern in Young's double slit experiment. (iii) In the diffraction due to a single slit experiment, the aperture of the slit is 3 mm. If monochromatic light of wavelength 620 nm is incident normally on the slit, calculate the separation between the first order minima and the 3rd order maxima on one side of the screen. The distance between the slit and the screen is 1.5 m. OR (i) A point object is placed in front of a double convex lens (of refractive index $n = n_2/n_1$ with respect to air) with its spherical faces of radii of curvature R_1 and R_2. Show the path of rays due to refraction at first and subsequently at the second surface to obtain the formation of the real image of the object. Hence obtain the Lens maker's formula for a thin lens. (ii) A ray of light is incident at an angle 45° at one face of an equilateral prism and passes symmetrically through the prism. Calculate : (a) the angle of deviation produced by the prism. (b) the refractive index of the material of the prism.		5

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